

GRANT NEILSON

Bioinformatician with six years of industry experience analyzing multiple NGS data types, with a specialism in whole genome sequencing (WGS). Highly proficient in R and Python, with a strong track record of developing analytical tools and packages for high-dimensional omics data. Passionate about building robust analytical workflows and contributing to open-source repositories to drive forward scientific research. An articulate and reliable individual with excellent communication skills and a strong track record of effective collaboration across multidisciplinary teams.



INDUSTRY EXPERIENCE

2026
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2023

Senior Computational Biologist

CoSyne Therapeutics

📍 London, UK

- Architected and deployed containerized bioinformatics infrastructure using Docker and AWS ECR, establishing scalable cloud-native solutions for multi-omic data analysis. Designed and implemented CI/CD pipelines for reproducible computational workflows, reducing deployment time by 60%.
- Lead developer and maintainer of multiple internal Python and R packages, establishing comprehensive testing frameworks, documentation standards, and code review processes. Implemented project templating system using Cruft, standardizing development practices across the organization.
- Engineered hyperscale cloud infrastructure on AWS for processing ~3 million single-cell sequencing samples, implementing automated alignment and QC pipelines. Optimized compute resource allocation and parallelization strategies, achieving 50% cost reduction and 30% performance improvement.
- Contributed to open-source NF-core Sarek pipeline (v3.4.1), implementing performance optimizations and workflow improvements. Developed custom Nextflow modules for variant calling and quality control processes.
- Established software engineering best practices including version control workflows, code review standards, unit testing frameworks, and comprehensive documentation. Mentored team members on software development methodologies and DevOps practices.

View this CV [online](#)

CONTACT

✉ grant.neilson5@gmail.com

🐙 [GitHub](#)

in [LinkedIn](#)

LANGUAGES

📊 R
🐍 Python
🔗 Bash
🔗 Nextflow

TECHNOLOGIES

🐙 GitLab/GitHub
🐳 Docker
aws aws
🖥 Azure

OPEN SOURCE & PUBLIC PROJECTS

🐙 [Sarek](#)
🐙 [bixverse](#)
🐙 [mainfoldsR](#)

Made with the R package [pagedown](#).

The source code is available on [GitHub](#).

Last updated on 2026-02-17.

2026
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2023

Senior Computational Biologist

CoSyne Therapeutics

📍 London, UK

- Drove biomarker discovery in glioblastoma by conducting WGS analysis on 80 primary tumour-derived cell lines, delivering comprehensive genetic characterisation (SNVs, SVs, CNVs). Integrated genomic profiles with CRISPRi screen data using advanced statistical models to uncover novel genetic dependencies and biomarker-target pairs.
- Designed and implemented the end-to-end bioinformatic analysis for high-throughput single-cell Mosaic drug screening, deploying hyperscale cloud infrastructure to run alignment and quality-control pipelines for ~3 million cells. Designed and implemented scalable pipelines for cell demultiplexing, sensitivity profiling of cell lines, and multi-omic biomarker identification.
- Executed single-cell CNV (sc-CNV) calling on proprietary perturb scRNA-seq data, identifying sub-clonal populations and contributing to the target identification platform.
- Recognized as a named contributor in version 3.4.1 of the NF-core Sarek pipeline. Optimised the running of the pipeline, achieving a 50% cost reduction and 30% decrease in computational time per sample.
- Internal R&D: Responsible for developing an internal container system using Docker and ECR (later handed over to engineering). Author and Lead developer on several internal R and Python packages with responsibilities for maintenance. Established best practices for code development and review. Aided in the development of a project templating system using cruft.

2023
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2020

Computational Biologist

Fios Genomics

📍 Edinburgh, UK

- Directed projects focusing on high-dimensional omics data types (WES, WGS, scRNA-seq, proteomics), delivering comprehensive analyses and reports for clients.
- Developed in-depth knowledge of statistical methodologies, mastering model selection and fitting techniques, including SVM, elastic net, random forest, and linear/logistic regression, while accounting for covariates.
- Conducted complex regression model analyses for biomarker discovery in bladder and lung cancer.
- Actively contributed to internal R&D by developing and maintaining specialized R packages, enhancing the team's analytical capabilities.
- Led client calls, effectively communicating complex analytical results, ensuring clarity and client understanding.

2020
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2019

Computational Biologist

Complex Diseases Epigenetics group, Exeter Medical School

📍 Exeter, UK

- Developed novel bioinformatics pipelines for Epigenome/Genome wide association studies, mQTL analysis and quality control of Illumina EPIC array data.
- Took the lead on a project performing EWAS and GWAS analysis in ADHD patients. This involved fitting linear and logistic regression models while accounting for covariates.
- Was selected for a secondment to New York to work at the Mount Sinai medical school as a visiting bioinformatician. This was unfortunately cancelled due to COVID-19.



EDUCATION

2019
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2015

MSCi Biology

University of Southampton

📍 Southampton, UK

- Grade : First Class Honors
- Winner of the British Society for Immunology Graduate of the year award.
- Recipient of the summer studentship from the Genetics Society (2017).

2015
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2013

School

Merchiston Castle Shool

📍 Edinburgh, UK

- A Levels: Biology, Chemistry, Maths



SELECTED PUBLICATIONS

2025

Multi modal single cell foundation models via dynamic token adaptation

BioRxiv

- Wenmin Zhao, AnaSolaguren-Beascoa, Grant Neilson, Regina Reynolds, Louwai Muhammed, Liisi Laaniste, Sera Aylin Cakiroglu
- Role: Data Preprocessing
- doi: <https://doi.org/10.1101/2025.04.17.649387>

2025

Detecting cell level transcriptomic changes of Perturb seq using Contrastive Fine tuning of Single Cell Foundation Models

BioRxiv

- Wenmin Zhao, AnaSolaguren-Beascoa, Grant Neilson, Regina Reynolds, Louwai Muhammed, Liisi Laaniste, Sera Aylin Cakiroglu
- Role: Data Preprocessing
- doi: <https://doi.org/10.1101/2025.04.17.649395>

2021

Novel epigenetic clock for fetal brain development predicts prenatal age for cellular stem cell models and derived neurons

Molecular Brain

- Leonard C. Steg, Gemma L. Shireby, Jennifer Imm, Jonathan P. Davies, Alice Franklin, Robert Flynn, Seema C. Namboori, Akshay Bhinge, Aaron R. Jeffries, Joe Burrage, Grant W. A. Neilson, Emma M. Walker, Leo W. Perfect, Jack Price, Grainne McAlonan, Deepak P. Srivastava, Nicholas J. Bray, Emma L. Cope, Kimberley M. Jones, Nicholas D. Allen, Ehsan Pishva, Emma L. Dempster, Katie Lunnnon, Jonathan Mill & Eilis Hannon

2020

Recalibrating the epigenetic clock: implications for assessing biological age in the human cortex

Brain

- Gemma L Shireby, Jonathan P Davies, Paul T Francis, Joe Burrage, Emma M Walker, Grant W A Neilson, Aisha Dahir, Alan J Thomas, Seth Love, Rebecca G Smith, Katie Lunnon, Meena Kumari, Leonard C Schalkwyk, Kevin Morgan, Keeley Brookes, Eilis Hannon, Jonathan Mill
- Role: Data Analysis and Alignment